

How Altered Time Scales Affect Motor Skill Acquisition in Virtual Reality

Category: Research

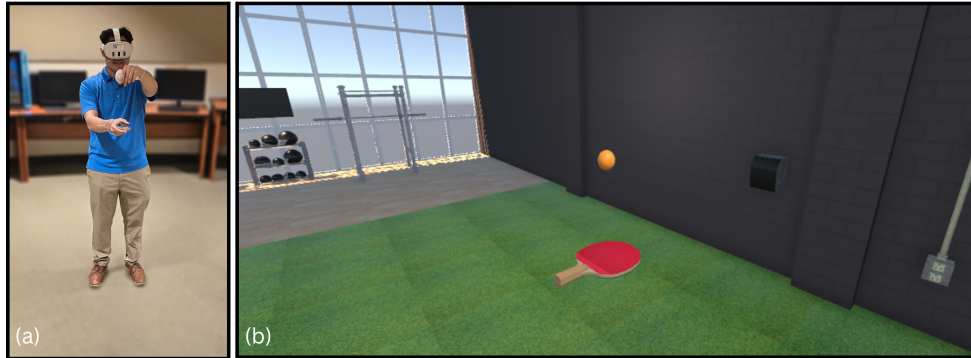


Figure 1: User performing the task from the third-person perspective in the (a) real world and (b) virtual recreation room. The left hand controller was used to spawn and release the ball for each attempt while the right controller served as the virtual paddle.

ABSTRACT

Virtual reality (VR) affords unique opportunities to manipulate training conditions in ways that are difficult to achieve in the real world. In this work, we investigate how temporally altered training affects motor skill acquisition in novices in VR. Participants performed a paddle-juggling task across pre-test, training, and post-test phases where they were randomly assigned one of three training paradigms: Slowed-down, Sped-up, or Gradually Sped-up. Temporal alterations were applied only during training, while performance before and after training was assessed identically in real-time dynamics. Results showed that slowed-down training altered behavior without improving performance, whereas both Sped-up and Gradually Sped-up training significantly improved post-test performance. The temporal schedules also produced distinct behavioral adaptations, indicating differences in how the task was performed. Although the accelerated schedules yielded comparable performance gains, Gradually Sped-up training was perceived more favorably. These findings highlight the importance of temporal structure in VR-based motor training and suggest that progressively increasing temporal demands may provide an effective approach for enhancing motor skill acquisition.

Index Terms: Virtual reality, motor learning, temporal manipulation, perception-action coupling.

1 INTRODUCTION

Virtual reality (VR) and broader extended reality (XR) technologies have emerged as powerful platforms for skill training, enabling learners to practice complex tasks in controlled, repeatable, and interactive environments. As VR and XR technologies have matured, they have been increasingly adopted in medicine [80], manufacturing [57], sports [56], and military training [5], where skill acquisition is critical for performance and safety. Many VR training applications focus on motor learning, which requires the coordination of perception and action through repeated practice and performance refinement [29]. By enabling systematic manipulation of training conditions and continuous collection of performance data, VR provides a unique environment for studying and facilitating motor skill acquisition. Indeed, research has demonstrated that VR-based training can improve learning outcomes and task performance [35].

Fine and gross motor skills represent two fundamental categories of human movement. Fine motor skills involve precise, coordinated actions of the hands and fingers, such as surgical suturing, tool manipulation, and component assembly, whereas gross motor skills rely on larger muscle groups to support actions such as running, climb-

ing, lifting, and maintaining balance [52]. Many real-world tasks require both types of movement to work together. For example, surgeons combine stable posture with precise instrument control [43], table tennis players coordinate whole-body movement with accurate racket positioning [36], and military personnel often transition between mobility-intensive actions and precision-dependent tasks such as weapon operation or combat casualty care [37]. These integrated motor skills are important in healthcare, sports, military, and industrial settings, where performance depends on both movement accuracy and physical coordination under dynamic conditions. Consequently, training methods must produce meaningful improvements in motor performance efficiently, enabling learners to achieve competency within practical time and resource constraints, and developing such approaches is helpful across many domains [79].

Prior research suggests that motor skill acquisition can be improved through training interventions that adapt task difficulty, provide targeted feedback, and structure practice conditions. Matching the difficulty of a task to a learner’s capabilities has been shown to improve balance-related motor learning [58], while other work shows that provisioning real-time posture feedback in VR can enhance skill acquisition by helping users correct movement errors during practice [55]. Similarly, Lammfromm and Gopher found that novice jugglers could successfully acquire and transfer skills to real-world performance when trained under progressively increasing temporal demands [30]. Together, these findings suggest that training effectiveness depends both on what learners practice and on the design of their practice conditions.

Among the factors that can be manipulated in VR, time is particularly promising because successful motor performance relies on precise coordination between perception and action. Prior work has explored temporal manipulation for athletic training, including table tennis through temporal cuing [76] and tennis through mediated-timescale learning [27]. However, these studies examined specific interventions within individual applications, leaving limited understanding of how different temporal manipulations compare against each other. This gap is especially important given evidence that motor timing can improve under accelerated conditions [19], suggesting that training at faster-than-normal speeds may produce unique learning benefits. A systematic investigation of temporal manipulation strategies could enhance VR-based motor skill training and provide broader insights into motor skill acquisition.

In this work, we address gaps in our understanding of temporal manipulation for motor skill training by evaluating how different temporally altered training paradigms affect skill acquisition in VR.

In a paddle-juggling task that integrates fine and gross motor skills with hand-eye coordination, we altered the timescale of a ball during training and examined its effects on performance, behaviors, and subjective perceptions. We conducted a study in which participants performed the task across three sequential phases: pre-test, training, and post-test, with the temporal manipulations applied only during training. Three training paradigms were evaluated: Slowed-down, Sped-up, and Gradually Sped-up. We present results showing that training paradigms with accelerated time flows improve motor skill acquisition, while slowed-down training paradigms are perceived more favorably. Apropos of these findings, progressively increasing time flow emerges as a promising approach for achieving performance gains while fostering a more favorable user experience, informing the design of immersive XR training systems.

2 RELATED WORK

2.1 Motor Task Training in Virtual Reality

Motor tasks can involve both fine movements using the smaller muscles of the hand for object manipulation and gross movements of the trunk and extremities for positioning, and a single task may require both [52]. Paddle juggling combines these demands in a repeated interceptive task, where actions must be coordinated with a moving ball while its trajectory is still unfolding. Research on table tennis shows that interceptive actions continue to be regulated during execution [5], while ball-catching studies demonstrate that contact timing depends on available optical information and can be influenced by prior knowledge about the moving object [8,38]. Juggling extends this problem across linked actions because each interception influences the trajectory that must be controlled next. Studies of cascade juggling show that improved performance is associated with increasingly stable temporal and spatial coordination [3,21]. Related work on rhythmic ball bouncing has likewise shown that performers can exploit dynamically stable paddle-ball interactions while applying active corrections when necessary [61,67,74]. Together, these findings characterize paddle juggling as a task requiring repeated regulation of both contact timing and the resulting ball trajectory.

VR provides a controlled environment in which motor skills can be practiced while maintaining control over the training task. VR has been used to train surgical manipulation [63], baseball batting [16], karate [56], and table tennis [48]. Transfer from virtual practice to real-world performance has been demonstrated in surgery and baseball [16,63], as well as in table tennis [48,54]. VR karate training has also produced learning comparable to video instruction [56]. More broadly, reviews of VR motor training highlight substantial variation in feedback, interaction techniques, training design, and outcome measures [44,53], including across ball-sport applications [49]. The extent to which training transfers beyond VR has therefore been argued to depend on how well the perceptual, cognitive, and motor features relevant to the real task are preserved [35].

Beyond reproducing a motor task, VR allows researchers to manipulate both the information provided during practice and the dynamics of the task itself. Visual, auditory, and haptic feedback can communicate movement features or action outcomes [65]; for example, trajectory feedback and real-time posture feedback have been used during virtual table-tennis training [55,69]. VR can also alter the demands of interceptive actions through changes in ball speed [25], trajectory shape and orientation [71], or the virtual representation of the user's end effector [72]. The mapping between physical and virtual movement can even be manipulated directly: delaying the virtual representation of racket motion during ball bouncing led participants to adjust their physical movement to recover an effective impact phase [51]. These capabilities make VR useful not only for practicing a motor task, but also for systematically changing the conditions under which that task is learned.

Juggling provides a particularly relevant example of such task manipulation. Immersive VR juggling has been used to support

learning and transfer to real-ball performance [28], while reduced virtual gravity has been used to slow the task during practice [1]. A mixed regimen combining real juggling with virtual progressive-speed and temporal-spatial emphasis practice produced real-ball performance comparable to real-only practice despite fewer physical sessions [30], while augmented feedback has been used to target throw height and timing [6]. More recently, progressively increasing virtual gravity has been reported to benefit early juggling relative to constant reduced gravity, although the reported group differences were descriptive rather than statistically reliable [9]; subsequent work on the same experiment decomposed early juggling performance into sequencing, prediction, and accuracy [10]. Related virtual object-control research has examined manipulated success experiences in VR kendama by adjusting invisible collision-detection boundaries and positive reinforcement [26], haptic interaction in mixed-reality juggling [4], and throwing tasks in virtual environments [24]. Building on this literature, we use one-ball paddle juggling as a repeated interceptive motor task and manipulate its temporal demands across distinct training schedules while preserving the underlying interaction and coordination requirements. This shifts the question from whether VR can alter motor practice to how temporal difficulty should be structured across practice, motivating the literature on training difficulty and progression discussed next.

2.2 Practice Difficulty and Training Progression

Practice difficulty can influence both immediate performance and subsequent motor learning, but greater difficulty is not inherently more beneficial. The challenge-point framework proposes that an appropriate level of difficulty depends on the interaction between task demands and the learner's current ability [17], with later extensions also emphasizing motivation and practice specificity [22]. The effects of practice conditions can further depend on task complexity, as principles observed for simple laboratory tasks do not always generalize to more complex skills [77]. This is evident in the use of practice support: reducing feedback can improve later retention by limiting dependence on external guidance [34,75], whereas frequent feedback can aid acquisition of more complex movements [78]. Similarly, reducing errors during early practice can help learners establish a movement pattern when the task is initially difficult [46]. Together, these findings suggest that useful practice difficulty depends on both the task and the learner's stage of learning.

Performance during practice also does not necessarily predict how well a skill will be retained or transferred. Schema theory proposes that experience across different movement parameters can support more generalizable motor representations [62]. Consistent with this idea, random practice can impair performance during acquisition relative to blocked practice while producing better subsequent retention and transfer [64]. However, these contextual-interference effects depend on factors such as task complexity and learner experience [41]. Thus, practice conditions that make a task easier or produce stronger immediate performance are not necessarily those that produce the strongest learning afterward.

These principles motivate training approaches in which difficulty changes over time rather than remaining fixed. Adaptive rehabilitation systems have adjusted task difficulty according to ongoing performance or changes in participant ability [11,58]. Progressive difficulty has also been applied to virtual juggling, where training began below the realistic temporal demands of the task and increased as learners met performance criteria, with practiced temporal-spatial components contributing to physical juggling performance [30]. Such work supports increasing task demands as learning progresses, but leaves open how a predefined progression compares with practice that consistently remains easy or difficult. In VR, temporal scaling provides a means to alter this dimension of difficulty while allowing the learner's physical movement to remain largely unchanged, motivating the work on time manipulation discussed next.

2.3 Time Manipulation in VR

VR allows the temporal progression of virtual events to be altered without requiring users' physical movements to change at the same rate. Full-body slow-motion systems show that this decoupling is not trivial. Slowing the surrounding environment alone need not measurably change users' temporal experience, and may additionally require that the visual representation of users' movements be scaled [59]. However, coupling virtual time flow directly to a user's real-world movement produces temporal recalibration that persists after exposure [2]. Similar techniques have been applied to motor training with rapidly moving objects. Virtual tennis systems have adjusted ball motion according to users' movement [18] and manipulated the timescale of forehand-volley practice [27]. While dynamic coupling can introduce persistent sensorimotor recalibration, these systems demonstrate that decoupling virtual event timing from physical movement allows the temporal demands of an interceptive task to be modified while preserving the learner's underlying action.

Temporal manipulation has been used as a training variable in VR sports. SPinPong incorporated temporal scaling alongside visual and haptic cues for difficult table-tennis returns, beginning each return below normal speed and ramping back to real time by the moment the ball reached the player [76]. Altered temporal dynamics have likewise been investigated in full-body VR ski training [45]. These approaches primarily use temporal scaling to reduce the initial demands of a task and then restore them as training progresses, suggesting that the rate at which virtual events unfold can itself be structured as part of practice. Manipulating virtual motion can also influence the subjective experience of time. Prior work has linked motion and apparent motion to changes in perceived duration [7, 73], and has examined time judgments in relation to task presentation and avatar embodiment [42, 70], self-motion and gameplay [31, 50], and manipulated object motion [32]. These findings indicate that temporal experience in VR depends on physical movement, virtual events, and interaction, rather than immersion alone.

Existing temporal-training systems have applied these manipulations mainly to discrete actions [27, 76], and have generally not evaluated learners under a common temporal condition after training. Whether practice held above real time is more demanding than practice below it is also unsettled: models of temporal scaling predict, and human psychophysics confirm, that timing precision can improve at faster speeds [19], so faster-than-normal dynamics may impose a qualitatively different demand rather than a uniformly greater one. Prior VR work provides limited evidence directly comparing practice held consistently below real time with practice held above real time, or with a progression that begins below real time and continues beyond it. It therefore remains unclear how these distinct temporal training schedules affect subsequent performance when learners are evaluated under the same real-time dynamics. Motivated by this gap, we developed a VR paddle-juggling system in which the temporal dynamics of the ball could be systematically scaled across training, enabling us to compare distinct temporal training schedules and evaluate their effects under common real-time dynamics.

3 SYSTEM DESCRIPTION

3.1 Apparatus

The IVE used for this study was developed in Unity 2022.3.22 using OpenXR and the Unity XR Interaction Toolkit. The application was rendered on a Meta Quest 3 HMD that was wirelessly connected (using the Meta Quest Air Link) to a computer equipped with an AMD Ryzen 7 7700 8-Core Processor, with 32 GB of RAM, and an NVIDIA GeForce RTX 5060 graphics card. Interaction was facilitated using two Quest 3 controllers.

3.2 Virtual Environment and Task Components

The ping pong paddle-juggling task described in section 4.1 was housed in a simple indoor virtual recreation room, featuring miscel-

laneous objects and equipment that were strategically included to increase immersion. The task was performed against a dark wall to make the ball and paddle appear salient. Realistic textures and colors were chosen for the ball and paddle to both increase the concordance with real-world ping pong equipment as well as to make them adequately contrast against the wall. The right controller served as a virtual table-tennis paddle, while the left controller allowed users to spawn and release a single virtual ball, identical to the contemporary XR table-tennis application Eleven Table Tennis [15] which has been used as a platform to study motor skill acquisition [48, 68]. The paddle followed the tracked pose of the right controller, while the ball was simulated using Unity's rigid-body physics, responding to gravity, paddle contact, and the temporal manipulations described in section 3.3. Brief text and auditory cues informed participants when a phase began or ended without introducing condition-specific visual information. The appearance and spatial arrangement of the environment remained unchanged throughout the experiment.

Pressing the left controller's trigger button resulted in a ball spawning at a predefined spawn-point affixed to the left controller, while the spawned ball remained under control (grasp) for as long as the trigger button was held pressed. Ball collisions were disabled while the ball was held to prevent contact before release. Upon release, the ball was launched using the measured velocity of the left controller obtained from the controller's rigid-body motion calculated over the preceding 20 frames (Unity default). Applying this scaled velocity allowed users to naturally initiate the task, either tossing or simply releasing the ball. Once released, users leveraged the paddle to perform the task, attempting to bounce the ball on the paddle repetitively. These interaction mappings remained constant throughout the experiment and the tuned parameters used to facilitate the task's mechanics are described in section 3.4. Paddle-ball contacts were algorithmically processed before being counted as valid bounces to avoid misattributing superior performance to grazing contacts, rolling contacts, and repeated collision events generated by the same physical impact. The specific thresholds chosen for bounce-validations and their descriptions are listed in table 1.

3.3 Timeflow Manipulation

The temporal progression of the virtual ball was varied by manipulating its timeflow scale τ in the training phase of the study (section 4.2). A value of $\tau = 1.0$ produced real-time ball dynamics while values below and above 1.0 slowed down and sped up the dynamics, respectively. The manipulation of this parameter affected the dynamics of the virtual ball without changing the HMD's refresh rate, the tracking of the virtual paddle, and the participant's physical movements. The paddle therefore continued to follow the position and orientation of the right controller in real time while the ball (when separated from the left controller) was the only virtual artifact affected by the manipulation of τ . This separation allowed the temporal demands of the virtual task to be altered without slowing or accelerating the participant's tracked hand movements. The three temporal training paradigms empirically evaluated in this work included a *Slowed-down*, a *Sped-up*, and a *Gradually Sped-up* condition. The specific time schedule used to realize each of these conditions is listed in section 4.2. Manipulations to this parameter τ modified both the gravitational acceleration and velocity of the virtual ball in the manner explicated below.

Let $\mathbf{g}_0 = (0, -9.81, 0)$ m/s² represent the gravitational acceleration used under real-time conditions. At timeflow scale τ , the application calculated the ball's gravitational acceleration $\mathbf{g}_\tau$ as:

$$\mathbf{g}_\tau = \tau^2 \mathbf{g}_0 \quad (1)$$

Scaling by τ^2 changed the magnitude of the gravitational acceleration applied to the ball. Whenever the active timeflow scale changed, the application also adjusted the ball's current velocity according to

$$\mathbf{v}_{\text{new}} = \frac{\tau_{\text{new}}}{\tau_{\text{previous}}} \cdot \mathbf{v}_{\text{previous}} \quad (2)$$

Together, these adjustments implemented timeflow by modifying the ball’s physical dynamics rather than by changing the rendering or playback speed of the virtual environment. For the Gradually Sped-up condition, the timeflow scale was updated evenly between training attempts, starting with scale used in the Slowed-down condition ($\tau = 0.65$) on the first attempt and ending with the scale used in the Sped-up condition ($\tau = 1.20$) on the last attempt. Thus, the timescale $\tau(n)$ at the n^{th} attempt in this condition was calculated as:

$$\tau(n) = 0.65 + \frac{n-1}{19}(1.20 - 0.65), \quad n \in \{1, \dots, 20\} \quad (3)$$

The resulting timeflow scale remained fixed for the entire duration of each attempt in this condition and was updated before every next attempt. Section 3.4 describes how the parameters used in the simulation were determined and the rationale behind their choices.

3.4 Pilot Testing and Parameter Tuning

In order to determine the appropriate parameters to be used in the simulation and experiment, a total of 27 pilot sessions were iteratively conducted. Each session lasted about half an hour and required participants to perform the paddle-juggling task with a defined set of parameter configurations. After performing the task, participants provided subjective feedback, answering questions pertaining to the naturalness of the simulation and ball-dynamics, intuitiveness of the interaction mappings, their fatigue levels, their perceived challenge in performing the task, and aspects that they felt warranted improvement. Data and feedback obtained from each of these pilot sessions were used to guide subsequent parameter tuning. Table 1 lists the finalized parameters used in the study.

The *bounce-validation parameters* were tuned to reject weak or incidental contacts that the ball made with the paddle, thereby preventing repeated collision events from the same impact from being counted separately and consequently belying performance. For the *experimental parameters*, the maximum number of bounces per attempt was selected to balance task challenge with the need to avoid fatigue from unusually long attempts. The number of attempts in each phase was selected to provide sufficient repeated measurements while limiting fatigue and maintaining the planned study duration. Training included more attempts than the pre-test and post-test to provide participants with greater exposure to their assigned timeflow condition and maximize experimental power. The two time scales used in the Slowed-down and Sped-up conditions were selected

Table 1: Overview of parameters. Light shading denotes bounce-validation thresholds and dark shading denotes study parameters.

Parameter	Description	Value
Bounce timing requirements	Minimum time between counted bounces and minimum time the ball must remain separated from the paddle before a new impact can be evaluated.	Contact: ≥ 0.12 s Off-paddle: ≥ 0.08 s
Post-contact separation	Minimum distance the ball must travel away from the paddle after contact for a bounce to be valid.	≥ 0.05 m within 0.25 s
Relative speed threshold	Minimum speed difference between the paddle and ball required for a valid bounce.	≥ 0.35 m/s
World speed threshold	Minimum ball speed at contact required for a valid bounce.	≥ 0.28 m/s
Maximum bounces per attempt	Maximum number of bounces allowed in one attempt.	15
Attempts per phase	Number of attempts completed in each study phase.	pre-test: 15 training: 20 post-test: 15
Timeflow floor and ceiling	Lowest and highest time scales applied to the ball during training.	$\tau_{\text{floor}} : 0.65$ $\tau_{\text{ceiling}} : 1.20$

to maximize power and ensuring that the task involved sufficient challenge while not being overly difficult or easy to be performable within a reasonable amount of time. These schedules also served as the first and last values of τ in the Gradually Sped-up condition, allowing us to evaluate whether progressively varying time flow over attempts affected performance, behaviors, and perceptions differently than fixed schedules. Prior work using altered ball dynamics in VR juggling also informed the selection of the timeflow range [9].

4 EXPERIMENT

4.1 Task

This experiment used a ping pong paddle-juggling task to evaluate different temporal training paradigms for a dynamic motor skill task requiring hand-eye coordination. Users were tasked with repeatedly bouncing a ball on a ping pong paddle, attempting to keep it airborne for as long as possible. Unlike tasks that involve a single interception event, this task requires several interceptions as well as motor control for one to successfully keep the ball in the air. Prior work has shown that although interceptive actions can be adjusted during movement, successful juggling depends on maintaining stable temporal and spatial coordination patterns across repeated cycles [3, 5, 21]. Along these lines, each time a user taps (i.e., bounces) the ball with the paddle, they are not only completing an interception but also influencing the ball’s subsequent trajectory which must henceforth be controlled and intercepted again. Consequently, the task requires the integration of fine motor control to precisely orient the paddle and execute strokes, gross motor movements to position the arm and body relative to the ball, and continuous hand-eye coordination to perceive, predict, and respond to the ball’s changing trajectory.

Users performed the task over three consecutive phases (section 4.2) in a virtual recreation room. The left controller was used to spawn and release the ball and the right controller served as the ping pong paddle (section 3.2). In each phase, users were given a predetermined number of attempts to perform the task. To initiate an attempt, users could simply spawn the ball using the left controller’s trigger button and release it by releasing the trigger. An attempt began only after the first valid paddle-ball contact because the task was intended to measure users’ skill in repeatedly bouncing the ball on the paddle rather than also measuring their efficacy in successfully releasing the ball onto the paddle. Thus, a ball that was released and dropped without making contact with the paddle did not constitute an attempt and was not included in data used for analyses. An attempt ended either when users lost control of the ball and it made contact with the floor, or when they managed to consecutively bounce the ball on the paddle 15 times, a threshold that was designated as signifying proficiency in the virtual paddle-juggling task based on data from pilot sessions. In most cases, attempts ended when the ball made contact with the floor, after which users could simply use the left controller to spawn and release a new ball to initiate the next attempt. If users successfully bounced the ball 15 times in an attempt, however, the ball disappeared and a message reading “Maximum Performance Reached” was displayed, indicating that they could proceed onto the next attempt. This sequence of events continued until attempts (eventually all phases) were completed.

4.2 Study Design

We employed a mixed factorial design, manipulating the type of temporal training users received (between-subjects) across three experimental conditions: (1) Slowed-down, (2) Sped-up, (3) Gradually Sped-up. Users were randomly assigned to one of these aforementioned conditions and performed a paddle-juggling task described in section 4.1 over three consecutive phases (within-subjects) namely pre-test, training, and post-test respectively. The temporal manipulation was only applied to the training phase, while both the pre and post-test were identical and free of temporal alterations (i.e., regular real-time dynamics), enabling comparisons between the methods

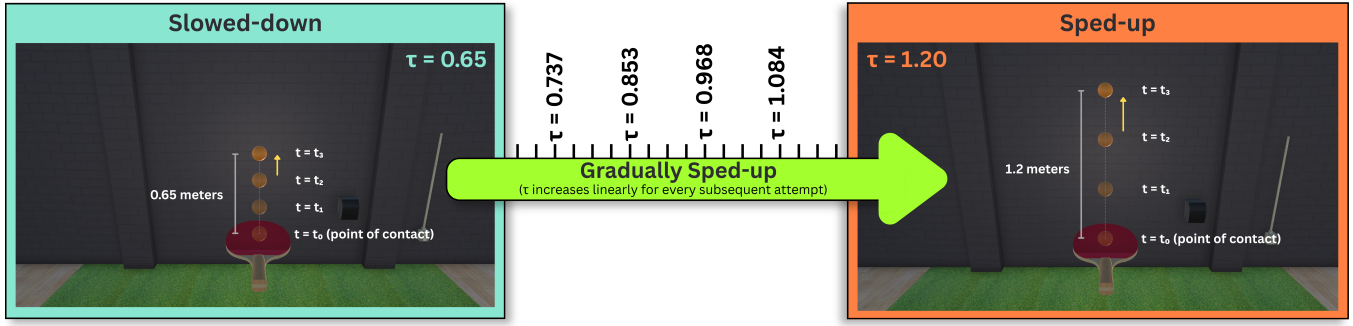


Figure 2: The left and right images show the time flow in the Slowed-down and Sped-up training paradigms where each condition's time scale τ was kept constant across all 20 attempts. For the paddle tapping the ball at a given hit velocity v , if the ball would normally travel 1 meter in real time (i.e., $\tau = 1$) in a given duration (from t_0 to t_3), the value of τ in each figure represents how much the ball would travel (0.65 meters and 1.20 meters respectively) in the same duration if hit with the exact same hit velocity. The Gradually Sped-up condition featured progressively increasing temporal scales from the Slowed-down's ($\tau = 0.65$) to the Sped-up's ($\tau = 1.20$) timescales over the course of 20 attempts.

in facilitating training. All users performed the task over the three phases for a predetermined number of attempts (15, 20, and 15 for pre-test, training, and post-test phases respectively), thus constituting a within-subjects factor. The number of attempts were determined based on content described in section 3.4. The training conditions (described subsequently) only differed in terms of the time scale applied to the ball while everything else was kept consistent.

Slowed-down: In this training paradigm, the ball moved slower than real-time, with τ set at 0.65 for all 20 attempts in the training phase.

Sped-up: In this training paradigm, the ball moved faster than real time, with τ set at 1.20 for all 20 attempts in the training phase.

Gradually Sped-up: In this training paradigm, the ball moved faster on every next attempt, starting at $\tau = 0.65$ in the first attempt and ending at $\tau = 1.20$ in the 20th attempt in the training phase.

4.3 Measures

Number of Bounces: In each attempt, the total number of valid bounces (contacts) between the paddle and the ball was computed and served as an operational measure of performance. This value ranged from one to fifteen with the latter serving as the predetermined threshold signifying proficiency. A higher value in this measure corresponds to better task performance.

Apex Distance: For every attempt, the highest vertical point (apex) reached by the ball was tracked on each bounce. The 3D euclidean distance between this apex point and the center of the paddle was computed for each bounce and averaged across all bounces in that attempt. This is a measure of behavior as it represents how far the ball is hit from the paddle not just vertically but also fronto-laterally.

Workload: Data on users' perceived levels of workload was measured using the NASA TLX questionnaire [20].

4.4 Research Questions and Hypotheses

This work's overarching aim was to answer the question: “How do temporally altered training paradigms affect motor skill acquisition in VR tasks that involve fine and gross motor skills?” Downstream of this, we were interested in studying how these alterations affect behaviors and perceptions associated with such tasks. We developed the following hypotheses that reflect prior work discussed in section 2.

H1: All 3 training paradigms will improve task performance.

H2: Simply repeating the task will improve performance.

H3: The training paradigms will differentially influence juggling behaviors, resulting in distinct post-test behavioral tendencies.

H4: Ball-juggling behaviors will predict performance in the task.

H5: In terms of workload, Slowed-down and Gradually Sped-up training will be more favorably perceived than Sped-up training.

Each training paradigm in this study is expected to improve post-test task performance relative to the pre-test baseline for different

reasons. The Slowed-down condition is expected to produce performance gains by allowing users to learn how to better coordinate their movement patterns with the ball, a component that prior research has shown to improve perception-action coordination in dynamic affordance tasks [33]. Sped-up training is expected to produce performance gains because training under more demanding conditions can enhance subsequent performance under normal conditions, a principle commonly employed in athletic training [12]. Similarly, Gradually Sped-up training is expected to improve performance by progressively exposing users to increased task demands and scaffolding task-difficulty increases over attempts. Repetition of the task is also expected to improve performance as it is generally associated with practice-related improvements in tasks that involve motor skills and hand eye coordination [60]. Beyond performance outcomes, the three training paradigms are expected to differentially influence users' ball-juggling behaviors because each exposes them to distinct temporal constraints that encourage different strategies for bouncing the ball. In turn, these behavioral tendencies are expected to predict performance because the way users regulate the ball's trajectory influences their ability to repeatedly intercept it and perform the task. With subjective perceptions, Slowed-down and Gradually Sped-up training are expected to be perceived more favorably than Sped-up training because they do not subject users to a high level of challenge at the outset but instead provide a more manageable progression of task demands, likely resulting in lesser workload and frustration while promoting more positive perceptions of performance.

4.5 Participants

An a priori power analysis using G*Power, assuming an effect size of 0.25, $\alpha = 0.05$, power $(1 - \beta) = 0.85$, and a correlation among repeated measures of 0.5, indicated that a minimum of 39 participants was required, with 13 participants per condition. Thus, a total of 40 participants were recruited for this Institutional Review Board (IRB)-approved study, with 13 allotted to the Slowed-down and Sped-up conditions, and 14 to the Gradually sped-up condition. The inclusion criteria for participation required that participants identify as novices in paddle/racquet sports. Each participant completed 50 attempts (see section 4.2) across the experiment resulting in a total of 2000 attempts for analyses. Participant ages ranged from 20 to 35 years old ($M = 24.49$, $SD = 3.82$); 15 of whom identified as female and the rest, male. All participants were right-handed, reported normal or corrected-to-normal vision, and normal motor functioning of their upper body. Overall, prior VR and task-relevant racquet sports experience did not significantly differ across conditions.

4.6 Procedure

Upon arrival at the laboratory, participants read and signed an informed consent form. After consenting to participate in the study,

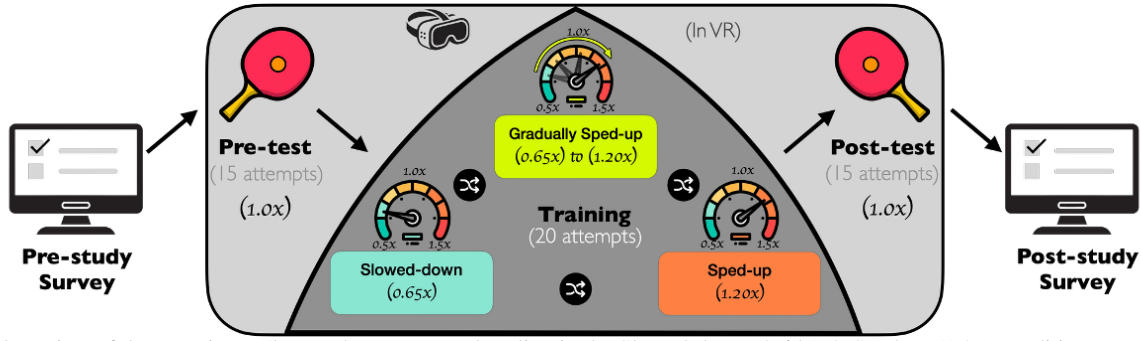


Figure 3: Overview of the experimental procedure. Temporal scaling in the Slowed-down (0.65x) & Sped-up (1.2x) conditions were kept fixed. The Gradually Sped-up condition featured progressively increasing temporal scales (0.65x-1.2x) that occurred only on each subsequent attempt.

users filled out a demographics questionnaire that included information about their backgrounds, prior experience with racket sports, table tennis, and VR. The experimenter then measured their interpupillary distance (IPD) and randomly assigned them to one of the three experimental conditions. Following this, the experimenter detailed the paddle-juggling task, further demonstrating how to use the controllers to perform it. Participants then donned the Meta Quest 3 (adjusted for their IPD) and it was ensured that the fit was snug and comfortable. They were given three practice attempts to familiarize themselves with the task and its mechanics. During these practice attempts, participants practiced spawning, releasing, and juggling the ball. Following this, participants proceeded to perform the juggling task over the three phases (described in section 4.2), filling out the NASA TLX questionnaire after the training phase to indicate their perceptions of workload associated with the temporal training paradigm. After completing a phase, participants were allowed to remove the HMD and take breaks. Before proceeding onto a subsequent phase, it was ensured that participants indicated feeling at ease, well rested, and not fatigued. Across participants, breaks lasted as long as 1.5 minutes. Upon completion of all three phases, they completed a post-study questionnaire and then engaged in a semi-structured interview probing into their experience. They were then debriefed and compensated for their time. On average, it took a participant about 45 minutes to complete the whole procedure.

5 RESULTS

Since a repeated measures design was used, variables had considerable nesting. As each participant completed 50 attempts, a portion of the variance in their responses can be attributed to a common source – the fact that the same participant made repeated attempts. Level 1 (within-participant) variables represent those that change from attempt to attempt. Level 2 (between-participant) variables represent those that change from participant to participant. Prior to conducting the analysis, the extent of nesting in the data was assessed by computing the intraclass correlation coefficient (ICC) from the null model for each dependent variable, using participant ID as a random effect. The ICC was found to be 0.34 for the number of bounces and 0.17 for the apex distance. Since this coefficient provides evidence of clustering by participant ID, and to properly account for variance between and within participants, linear mixed effects models were used to analyze each dependent variable [23, 47].

Since the number of bounces is a count variable, performing a Poisson regression is ideal in this case. However, while performing Poisson regression, if the dispersion parameter ϕ is greater than 1 (overdispersion), a negative binomial regression can be used [13]. In this data, since overdispersion was detected for the models with only the fixed effects, negative binomial regression was used to fix overdispersion. Linear mixed effects models were created to test the effects of the independent variables on the apex distance.

For each analysis, an initial main effects model was run such that the main effects were included in the analysis all at once. Results

for each main effect is reported from the initial main effects model. Analysis of two-way and three-way interactions was done by adding each interaction term to the main effects model separately. That is, separate models were run to obtain the results for each two-way and three-way interaction. All models also included a random effect of participant ID. Effect sizes for each fixed effect are presented as the change in R^2 (proportion of variance explained) comparing the model that includes the effect and the same model with the effect removed. The resulting sr^2 (semi-partial r^2) is the percentage of variance uniquely accounted for by the fixed effect [66].

5.1 Comparing Performance in Pre-test & Post-test

A negative binomial regression was conducted to assess the effects of condition, phase, and attempt number on the number of bounces. This model with only the main effect (AIC = 5363.9, $df = 7$) offered a significantly better fit than the null model (AIC = 5369.2, $df = 3$), $\chi^2 = 13.25$, $p = 0.01$. It explained 27.9% of the variance in number of bounces (conditional $R^2 = 0.279$, marginal $R^2 = 0.008$).

There was a significant effect of phase on number of bounces, $\chi^2(1, N = 1200) = 4.87$, $p = 0.027$, $sr^2 = 0.003$. Number of bounces was significantly greater in the post-test ($M = 4.27$, $SE = 0.274$) compared with the pre-test ($M = 3.96$, $SE = 0.255$), $z = -2.21$, $p = 0.03$. There was also a significant effect of attempt number on number of bounces, $\chi^2(1, N = 1200) = 8.07$, $p = 0.005$, $sr^2 = 0.005$. For every unit increase in attempt number, the expected number of bounces increased by a factor of 1.011. There was no effect of condition on number of bounces.

There was a significant interaction between condition and phase, $\chi^2(2, N = 1200) = 7.63$, $p = 0.022$, $sr^2 = 0.004$ (see Fig. 4a). When testing simple effects, participants in the gradually sped-up condition had significantly more bounces during the post-test ($M = 4.46$, $SE = 0.48$) than the pre-test ($M = 3.84$, $SE = 0.42$), $z(420) = 2.85$, $p = 0.004$. Similarly, participants in the sped-up condition had significantly more bounces during the post-test ($M = 4.46$, $SE = 0.50$) than the pre-test ($M = 4.00$, $SE = 0.45$), $z(390) = 2.01$, $p = 0.044$. However, in the slowed-down condition, the number of bounces did not differ between the pre- and post-test phases.

Phase was a significant moderator of the effect of attempt number on the number of bounces, $\chi^2(1, N = 1200) = 7.31$, $p = 0.007$, $sr^2 = 0.004$. As shown in Fig. 4b, a test of simple slopes revealed that for the post-test, the slope for attempt number was positive and significantly different from zero ($B = 0.02$, $SE = 0.007$, $z = 2.13$, $p = 0.03$), while the pre-test slope was non-significant ($B = -0.05$, $SE = 0.008$, $z = -0.646$, $p = 0.52$).

5.2 Effect of Repetition on Performance

On the pre-test data, a negative binomial regression was conducted to assess the effect of attempt number on the number of bounces. This model with only the main effect (AIC = 2633.3, $df = 4$) did not offer a significantly better fit than the null model (AIC = 2631.3, $df = 3$), $\chi^2 = 0.009$, $p = 0.92$. This indicates that the number of

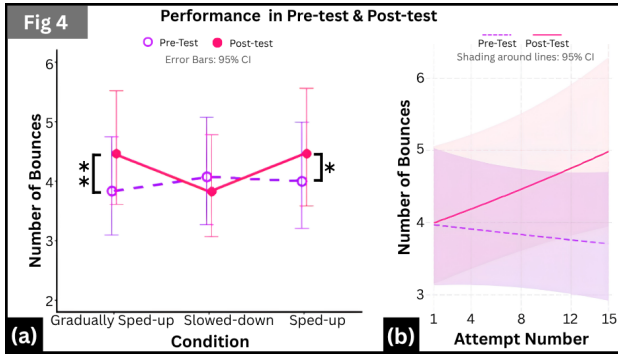


Figure 4: Effect of training type on performance.

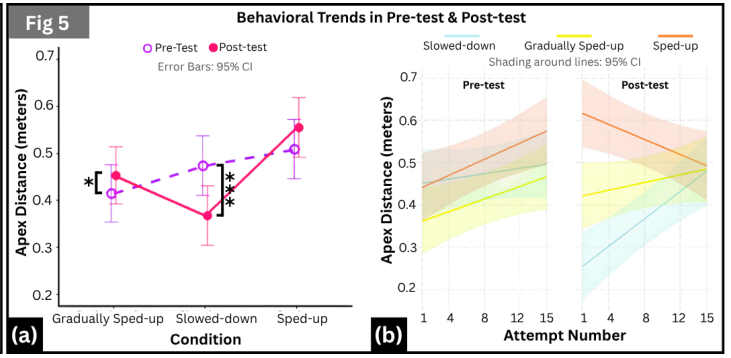


Figure 5: Effect of training type on behavior.

bounces did not improve with repeated attempts in the pre-test phase.

5.3 Comparing User Behavior in Pre-Test and Post-Test

A linear mixed-effects model was run to assess the effects of condition, phase, and attempt number on apex distance. This model with only the main effects ($AIC = -94.05$, $df = 7$) offered a significantly better fit than the null model ($AIC = -81.23$, $df = 3$), $\Delta\chi^2(4) = 20.82$, $p < 0.001$. It explained 19.7% of the variance in apex distance (conditional $R^2 = 0.197$, marginal $R^2 = 0.047$).

We found a significant effect of condition on apex distance, $F(2, 37) = 4.05$, $p = 0.026$, $sr^2 = 0.038$. Apex distance was significantly greater in the sped-up ($M = 0.532$, $SE = 0.03$) than in the slowed-down condition ($M = 0.421$, $SE = 0.03$), $t = -2.59$, $p = 0.035$. However, it was not significantly different between the sped-up and gradually sped-up conditions ($M = 0.434$, $SE = 0.03$), nor between the slowed-down and gradually sped-up conditions. There was also a significant effect of attempt number on apex distance, $F(1, 1158) = 12.73$, $p < 0.001$, $sr^2 = 0.008$. For each successive attempt, expected apex distance increased by 0.005 units.

There was a significant interaction between condition and phase, $F(2, 1153) = 14.94$, $p < 0.001$, $sr^2 = 0.019$ (see Fig. 5a). When testing simple effects, participants in the gradually sped-up condition had significantly greater apex distance during the post-test ($M = 0.454$, $SE = 0.031$) than during the pre-test ($M = 0.415$, $SE = 0.031$), $t(420) = 2.02$, $p = 0.045$. Participants in the slowed-down condition had significantly lower apex distance during the post-test ($M = 0.368$, $SE = 0.032$) than during the pre-test ($M = 0.474$, $SE = 0.032$), $t(390) = -4.917$, $p < 0.001$. However, in the sped-up condition, apex distance did not differ between the pre- and post-test.

Condition significantly moderated attempt number's effect on apex distance, $F(2, 1153) = 3.274$, $p = 0.038$, $sr^2 = 0.004$. As seen in Fig. 5b, a test of simple slopes revealed that the gradually sped-up condition's slope for attempt number was positive and different from zero ($B = 0.007$, $SE = 0.003$, $t = 2.555$, $p = 0.011$). Similarly, the slowed-down condition's slope was positive and significantly different from zero ($B = 0.011$, $SE = 0.003$, $t = 3.703$, $p < 0.001$). However, the sped-up condition's slope was not different from zero ($B = 0.002$, $SE = 0.003$, $t = 0.585$, $p = 0.559$).

There was a significant three-way interaction between condition, phase, and attempt number, $F(2, 1151) = 9.18$, $p < 0.001$, $sr^2 = 0.011$. As shown in Fig. 5b, a test of simple slopes revealed that for the pre-test, the slope for attempt number was positive for all three conditions. For the gradually sped-up ($B = 0.008$, $SE = 0.004$, $t = 2.131$, $p = 0.03$) and sped-up conditions ($B = 0.010$, $SE = 0.004$, $t = 2.613$, $p = 0.009$), the slopes were significantly different from zero, while for the slowed-down condition, the slope was not significantly different from zero ($B = 0.003$, $SE = 0.004$, $t = 0.884$, $p = 0.377$). For the post-test, the slope for attempt number was positive for the gradually sped-up and slowed-down conditions, but negative for the sped-up condition. For the sped-up condition ($B = -0.009$, $SE = 0.004$, $t = -2.427$, $p = 0.015$) and the slowed-

down condition ($B = 0.016$, $SE = 0.004$, $t = 4.413$, $p < 0.001$), the slopes were significantly different from zero, while for the gradually sped-up condition, the slope was not significantly different from zero ($B = 0.005$, $SE = 0.004$, $t = 1.294$, $p = 0.196$).

5.4 Performance in Training Phase

A negative binomial regression was conducted to assess the effects of condition and attempt number on the number of bounces in the training phase. This main effects only model ($AIC = 3915.8$, $df = 6$) offered a significantly better fit than the null model ($AIC = 3966.1$, $df = 3$), $\chi^2 = 56.3$, $p < 0.001$. It explained 52.4% of the variance in number of bounces (conditional $R^2 = 0.524$, marginal $R^2 = 0.354$).

There was a significant effect of condition on the number of bounces, $\chi^2(2, N = 800) = 69.19$, $p < 0.001$, $sr^2 = 0.372$. The number of bounces was significantly greater in the slowed-down condition ($M = 8.5$, $SE = 0.919$), $z = -3.001$, $p = 0.008$, and the sped-up condition ($M = 2.34$, $SE = 0.266$), $z = 8.23$, $p < 0.001$. The number of bounces was also significantly greater in the gradually sped-up condition than in the sped-up condition, $z = 5.403$, $p < 0.001$. There was also a significant effect of attempt number on number of bounces, $\chi^2(1, N = 800) = 16.62$, $p < 0.001$, $sr^2 = 0.009$. For every unit increase in attempt number, the expected number of bounces was reduced by a factor of 0.985.

Condition was found to be a significant moderator for the effect of attempt number on the number of bounces, $\chi^2(2, N = 800) = 152.77$, $p < 0.001$, $sr^2 = 0.036$. As seen in 6a, a test of simple slopes revealed that for the gradually sped-up condition, the slope for attempt number was negative and significantly different from zero ($B = -0.07$, $SE = 0.006$, $z = -12.22$, $p < 0.001$), while for the slowed-down condition, the slope was positive and significantly different from zero ($B = 0.023$, $SE = 0.005$, $z = 4.58$, $p < 0.001$). However, for the sped-up condition, the slope was not different from zero ($B = -0.001$, $SE = 0.008$, $z = -0.09$, $p = 0.928$).

5.5 User Behavior in Training Phase

A linear mixed effects model was run to assess the effects of condition and attempt number on the apex distance. This model with only the main effects ($AIC = 228.527$, $df = 6$) offered a significantly better fit than the null model ($AIC = 260.275$, $df = 3$), $\Delta\chi^2(3) = 37.748$, $p < 0.001$. It explained 40.8% of the variance in apex distance (conditional $R^2 = 0.408$, marginal $R^2 = 0.071$).

There was a significant effect of attempt number on apex distance, $F(1, 759) = 33.554$, $p < 0.001$, $sr^2 = 0.025$. For each successive attempt, the expected apex distance reduced by 0.009 units. There was no significant effect of condition on apex distance.

Condition significantly moderated the effect of attempt number on apex distance, $F(2, 757) = 54.374$, $p < 0.001$, $sr^2 = 0.069$. As shown in figure 6b, a test of simple slopes revealed that for the gradually sped-up condition ($B = -0.010$, $SE = 0.003$, $t = -3.767$, $p < 0.001$) and the slowed-down condition ($B = -0.029$, $SE =$

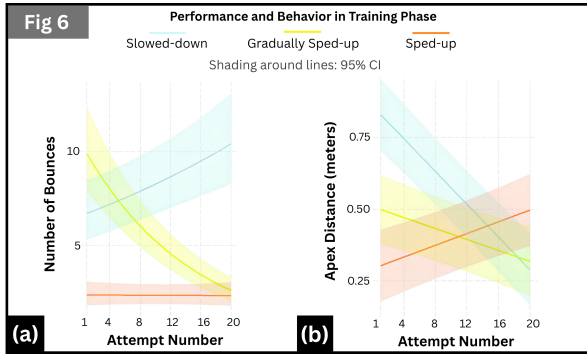


Figure 6: Performance and behaviors during training.

0.003, $t = -10.855$, $p < 0.001$), the slopes for attempt number were negative and significantly different from zero. However, for the sped-up condition, the slope was positive and significantly different from zero ($B = 0.01$, $SE = 0.003$, $t = 3.892$, $p < 0.001$).

5.6 Predicting Performance Based on User Behavior

On the pre- and post-test data, a negative binomial regression was run to assess the effect of apex distance and the total distance traveled by the ball, on the number of bounces. This model with only the main effects ($AIC = 5079.3$, $df = 5$) offered a significantly better fit than the null model ($AIC = 5369.2$, $df = 3$), $\chi^2 = 293.91$, $p < 0.001$.

There was a significant effect of apex distance on the number of bounces, $\chi^2(1, N = 1200) = 57.833$, $p < 0.001$, $sr^2 = 0.056$. For every unit increase in the apex distance, the expected number of bounces almost halved by a factor of 0.519. There was a significant effect of total distance traveled by the ball as well on the number of bounces, $\chi^2(1, N = 1200) = 844.85$, $p < 0.001$, $sr^2 = 0.241$. For every unit increase in the total distance traveled by the ball, the expected number of bounces improved by a factor of 1.072.

5.7 Subjective Perceptions of Workload

We compared the training conditions on each NASA TLX dimension using Kruskal-Wallis tests. When condition's effect was significant, post hoc comparisons were conducted using Wilcoxon rank-sum tests with Holm correction for multiple comparisons. There were no significant effects of condition on mental demand, physical demand, or overall workload. While condition's effect was significant on perceived effort, $\chi^2(2) = 6.34$, $p = .040$, $\eta_p^2 = 0.117$, no pairwise comparison reached statistical significance with Holm correction.

Perceived temporal demand differed significantly across conditions, $\chi^2(2) = 11.50$, $p = .003$, $\eta_p^2 = 0.257$. Pairwise comparisons indicated that the slowed-down condition ($Mdn = 1$) experienced significantly lower temporal demand than both the sped-up condition ($Mdn = 2.5$, $p = .021$) and the gradually sped-up condition ($Mdn = 2$, $p = .004$). No other pairwise differences were significant.

Perceived performance also differed significantly across conditions, $\chi^2(2) = 20.66$, $p < .005$, $\eta_p^2 = 0.504$. Pairwise comparisons revealed that the perceived performance in the slowed-down ($Mdn = 8$) was significantly higher than the sped-up ($Mdn = 2$, $p < .001$) and gradually sped-up ($Mdn = 5$, $p = .006$) conditions. The perceived performance in the gradually sped-up condition was also significantly higher than the Sped-up condition ($p = .006$) (Fig. 7a).

Perceived frustration differed significantly across conditions, $\chi^2(2) = 10.98$, $p = .004$, $\eta_p^2 = 0.243$. Pairwise comparisons indicated that slowed-down training ($Mdn = 1$) elicited significantly lower frustration than sped-up training ($Mdn = 4$, $p = .005$) and marginally lower frustration than gradually sped-up training ($Mdn = 2.5$, $p = .05$) (figure 7b). No other differences were found.

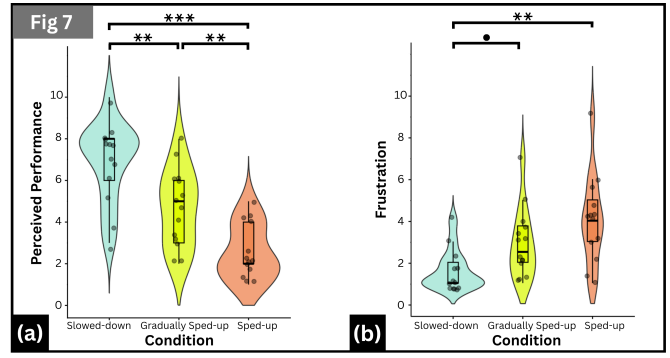


Figure 7: Perceptions of workload across training conditions.

6 DISCUSSION

The statistical analyses of performance across the pre and post-test phases shed light on the efficacy of the three training paradigms, also informing us about whether simply repeating the task made users more effective at keeping the ball airborne. With respect to training efficacy, we found a significant interaction effect (i.e., phase by condition), suggesting that the changes in performance from pre-test to post-test depended on the training paradigm participants underwent (section 5.1). Along these lines, training in the Sped-up and Gradually Sped-up conditions significantly improved performance from pre-test to post-test, while slowed-down training did not produce such performance gains (figure 4a). These results offered only partial support for **H1**, suggesting that temporally altered training for dynamic motor skill tasks may facilitate improvement when time flow is increased but not necessarily when decreased. The Gradually Sped-up training's efficacy being comparable to Sped-up training further indicates that even brief and recent exposure to challenging training conditions may be sufficient to achieve the training benefits observed by engaging in extended practice under such conditions (since only the last 7 attempts of this training phase featured $\tau > 1$). Overall, the benefits of faster time-flow training in such tasks align with practices commonly adopted in athletic training, wherein athletes practice under slightly more demanding conditions to enhance subsequent performance under standard conditions [12, 39]. Analogously, these findings also match experiences of watching videos at increased playback speeds, wherein returning to normal speed after adapting to the faster pace often makes movement feel comparatively slower and easier to process [14]. A caveat worth noting is that these trends may be specific to the time flows (i.e., τ values) investigated in our work and need not readily apply to other schedules. Along these lines, more subtle slowed-down training conditions (e.g., $\tau = 0.9$) may produce performance gains by allowing users to better learn coordination patterns that better transfer to realtime conditions (i.e., $\tau = 1$) while excessively sped-up conditions may worsen subsequent performance due to frustration that ensues from being unable to adapt. More research is needed to qualify how other points on the temporal training continuum affect performance.

We expected that repetition would improve task performance but did not find evidence of this effect as noted in section 5.2 which describes how performance changed over attempts in the pre-test. Since the post-test was performed after users underwent a temporally altered training phase, the effect of repetition was analyzed based on only the pre-test data to purely quantify (i.e., without any training) how attempting the task over and over again affected users' ability to perform the task. Since performance did not significantly improve with simple repetition (pretest line in figure 4b), **H2** was not supported. Notwithstanding the lack of evidence supporting task-repetition as a means to improve performance, prior research has consistently shown that dynamic motor skill acquisition in novices is facilitated by repeated practice of the target skill [40]. It thus appears that the number of attempts employed in our study may have been

insufficient to elicit the expected performance gains from repeating the task, likely requiring more repetition for such gains to emerge.

In terms of behaviors, analyses revealed that each temporal training condition differentially affected the manner in which users attempted to subsequently bounce the ball on the paddle. Users assigned to the Slowed-down and Gradually Sped-up conditions significantly changed the way they bounced the ball such that its distance from the paddle at the apex position (i.e., apex distance) varied from pre-test to post-test (figure 5a). Sped-up training, however, did not produce such behavioral shifts in terms of the apex distance maintained. Figure 5b further shows how users from the three conditions modified their juggling behaviors over attempts, across the pre and post-test phases. As seen in this figure, the pre-test behaviors of users assigned to the three conditions are largely similar, indicating comparable levels of baseline behavioral tendencies in terms of the way the ball was bounced over attempts. The post-test behaviors of users who underwent Gradually Sped-up training remained largely similar to the pre-test, indicating that this paradigm did not significantly alter behavior. In contrast, users across the other two conditions changed their juggling behaviors in the post-test. Along these lines, undergoing slowed-down training made users maintain a lower apex distance at first and then increase it over attempts while Sped-up training did the exact opposite (i.e., made users maintain a higher apex distance at first and subsequently reduce it over attempts).

These post-test behaviors can be explained by taking into consideration the fact that users previously performed the task in the training phase. Analyses of performance in this phase produced intuitive results, in that performance was best in the less challenging Slowed-down training, worst in the more challenging Sped-up training, and progressively declined as difficulty increased in the Gradually Sped-up training (figure 6a). Behaviorally, the slowed-down training enabled users to hit the ball less far away from the paddle (i.e., less apex distance), making them recalibrate (over attempts) from previously having performed the task in real time (i.e., in the pre-test). Similarly, the Sped-up schedule incentivized users to bounce the ball further away from the paddle (i.e., more apex distance) to meet this condition's increased challenges, showing a similar recalibrative trend (figure 6b). Thus, the differences in post-test behaviors between the conditions supports **H3**, reflecting the training paradigms' differences in terms of the behaviors they promote and the recalibrative behaviors subsequently required to successfully perform the task in real-time.

Behavioral tendencies in the way the ball was bounced also predicted performance significantly (section 5.6), thereby offering support for hypothesis **H4**. We found that increasing the apex distance when bouncing the ball significantly worsened performance, reducing the number of bounces subsequently expected. In contrast, increasing the total distance traveled by the ball improved performance. One possible explanation is that increasing the distance traveled by the ball provides users with additional time to position the paddle for the next interception while maintaining a lower apex distance could allow users to reach the anticipated interception point more quickly and reliably. Although these strategies may initially appear contradictory, they suggest that effective performance in this task involves generating sufficient ball flight time while avoiding unnecessarily far apices, potentially requiring follow-through movements of the paddle to reduce its distance to the ball while still allowing the ball to travel a sufficiently long path.

Analyses of subjective perceptions of workload associated with the different temporal training paradigms help add nuance to the discussion of their efficacies. As noted previously, we found that training with increased time flow schedules (i.e., either progressively or constantly sped-up) significantly improved task performance while slowed-down training did not. Despite the latter not producing performance gains, it was more favorably perceived in terms of workload, inducing lower levels of temporal demand and frustration,

while promoting higher levels of self-perceived performance (section 5.7, figure 7). Notwithstanding performance gains being comparable across the other two conditions, users reported significantly lower perceived performance following Sped-up training than Gradually Sped-up training (figure 7a), understandably because it was challenging at the outset. While both paradigms were perceived as more frustrating than slowed-down training (figure 7b), frustration was lower with Gradually Sped-up than Sped-up training. Although this difference did not reach statistical significance, the overall trends suggest that gradually introducing challenge may promote a more positive subjective training experience. These results offer considerable support for hypothesis **H5** and suggest that scaffolding an increased training timescale offers advantages beyond performance improvements alone. Consequently, designers of VR training experiences should consider progressively increasing timeflows given the balance it offers between training efficacy and user experience.

7 SCOPE AND LIMITATIONS

This work focused on understanding how temporally altered training schedules affect motor skill acquisition in VR. Accordingly, we compared three temporal training paradigms without including a conventional real-time training condition. Although this limits direct comparisons against unaltered training, the pre-test data allowed us to assess the effects of repetition alone. Notwithstanding our analyses providing insights into the efficacy of simple repetition, this work's scope was primarily to understand the relative efficacy of altered time scales for training. Furthermore, the temporal manipulations investigated here were restricted to two fixed timeflow scales ($\tau = 0.65$ & $\tau = 1.20$), along with a progression between them. Our findings may be limited to these schedules and should not be interpreted as applying uniformly across the continuum of temporal alterations, as more subtle or extreme manipulations may yield different tradeoffs between task difficulty, behavioral adaptation, and learning. It is also worth noting that we evaluated novices' performance solely within VR and did not assess subsequent skill-transfer to the real-world. Although virtual motor skill training has been shown to transfer to the real-world, the extent to such transfer holds true for temporally altered training paradigms remains open for investigation. Finally, while paddle juggling requires fine motor control, gross motor movements, and hand-eye coordination, it remains a relatively simple task. Consequently, our findings may not generalize to more dexterous tasks or ones that involve more complex motor skills. For example, tasks with multi-step movement sequences like cascade juggling may benefit from slowed-down training because reduced temporal demands could help learners acquire coordination patterns before attempting them at real-time speed.

8 CONCLUSION AND FUTURE WORK

We empirically evaluated how temporal training manipulations affect motor skill acquisition in novices in virtual reality. Participants performed a paddle-juggling task under slowed-down, sped-up, or gradually sped-up training conditions. Temporal alterations were applied only during training, whereas performance before and after training was assessed under unaltered real-time dynamics. Results showed that slowing time altered behavior without improving performance, whereas both sped-up and gradually sped-up manipulations improved motor skill performance. Although the latter two yielded comparable performance gains, gradually sped-up training was perceived more positively, suggesting that it may be a preferable approach for enhancing motor skill acquisition in VR.

This work raises several interesting follow up questions. How does temporally altered training in VR affect real world task performance? Would similar trends hold if progressive time scaling was applied over time instead of attempts? How does adaptive temporal scaling compare to strictly progressive time scaling? In future work, we aim to answer such questions and call for researchers to investigate what other implications altered time training entails.

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